

# UMER SIDDIQUI

## Backend & AI Systems Engineer | System Design

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[github.com/UmerSidiqui](https://github.com/UmerSidiqui) • [umersidiqui.github.io/Portfolio](https://umersidiqui.github.io/Portfolio)

## PROFESSIONAL SUMMARY

Backend and AI Systems Engineer with production experience building scalable, low-latency LLM-powered applications — agentic workflows, RAG pipelines, and enterprise voice AI — from architecture through cloud deployment. Strong grounding in core system design (API architecture, database selection, caching, load balancing, authentication/security) combined with applied AI infrastructure experience (model serving, orchestration, LLM Ops, cost and latency trade-offs). Comfortable owning a system end-to-end: designing the architecture, writing the backend, and reasoning about how it behaves at scale. Strong Data Structures & Algorithms fundamentals, applied through consistent problem-solving practice.

**Core Stack:** Python · FastAPI · Node.js/NestJS · TypeScript · PostgreSQL · pgvector · Redis · OAuth 2.0/JWT · Docker · AWS · GCP · CI/CD · LangGraph · RAG · System Design · DSA

## PROFESSIONAL EXPERIENCE

### Senior AI Engineer | Saynt AI 06/2025 – Present · Remote

- Owned end-to-end backend development for the voice agent dashboard and a dental practice management (PMS) platform — REST APIs, business logic, and authentication built in NestJS/TypeScript, deployed on AWS.
- Implemented authentication from the ground up, including Google OAuth login, across NestJS and FastAPI services.
- Set up the CI/CD pipeline for the NestJS backend using GitHub Actions — merges to main auto-deploy to AWS, removing manual release steps.
- Designed agentic workflows with LangGraph and RAG pipelines with webhook-based tool-calling, integrating Twilio telephony and PMS backends (OpenDental, Dentrix) for low-latency, context-aware voice interactions.

### Jr. AI Developer | Erly Stage Studios 10/2024 – 06/2025 · Remote

- Built the full backend for the WhatsApp Meeting Scheduling Agent in FastAPI — webhook handling, agent orchestration (LangGraph + OpenAI), and state management.
- Deployed the service on GCP with Firestore as the primary data store; collaborated with an Anthropic engineer on system architecture and scalability decisions.
- Built a product Q&A chatbot API in Flask over scraped product data in PostgreSQL, combining direct SQL queries with pgvector similarity search for hybrid structured + semantic retrieval; handed off the API for frontend integration into the webapp.

### AI & Data Analytics Intern | NCL NEDUET 04/2024 – 10/2024 · Karachi

- Delivered AI model development, EDA, and ML optimization projects with real-world business applications.
- Built LocoBot, a lab assistant robot combining real-time face recognition with a RAG chatbot for booking meetings with faculty; used directly by students visiting the lab.

## KEY PROJECTS

### Final Year Project — Novel Actor-Critic Algorithm for Reinforcement Learning

Reinforcement Learning · Actor-Critic · PyTorch · OpenAI Gym

Proposed a modification to the critic function in standard Actor-Critic architectures, achieving state-of-the-art performance surpassing prior baselines across multiple OpenAI Gym environments.

### Enterprise Voice AI Agent

ElevenLabs · Twilio · LangChain · RAG · PMS

Intelligent appointment scheduler with RAG context retrieval, webhook tool-calling, and real-time telephony integration, designed for low-latency production use.

### LocoBot — Face Recognition & Faculty Meeting Assistant

DeepFace · VGG-Face · Flask · SQLite · RAG

Lab robot that recognizes students in real time via a Flask API backend and uses a RAG chatbot to help them book meetings with faculty members — deployed for live use by students in the lab.

### Violence Detection System

YOLOv8 · Computer Vision · Flask

Retrained YOLOv8 for real-time video analysis with streaming annotated frame alerts. Detects violent activity in the video feed, extracts the relevant clip, and can forward it to officials for review.

## EDUCATION

**B.E. Electrical Engineering** — NED University of Engineering & Technology · CGPA: 2.5/4.0 · 02/2022 – 08/2026

**Higher Secondary (Intermediate)** — Bahria College Karsaz · BIEK Board · 996/1100 marks · 2020 – 2022

**Secondary School (Matriculation)** — CAA Model School and College, Karachi · BSEK Board · 755/850 marks · 2019 – 2020